

Google's carbon footprint balloons in its Gemini AI era

Posted by u/Balance- • 0 points • 22 comments

__Despite the company's climate goals, Google's greenhouse gas emissions are on the rise thanks to AI.__

Google's greenhouse gas emissions have ballooned, according to the company's latest environmental report, showing how much harder it'll be for the company to meet its climate goals as it prioritizes AI.

Google has a goal of cutting its planet-heating pollution in half by 2030 compared to a 2019 baseline. But its total greenhouse gas emissions have actually grown by 48 percent since 2019. Last year alone, it produced 14.3 million metric tons of carbon dioxide pollution — a 13 percent year-over-year increase from the year before and roughly equivalent to the amount of CO2 that 38 gas-fired power plants might release annually.

The jump in planet-heating pollution primarily comes from data center energy use and supply chain emissions, according to Google's environmental report. Data centers are notoriously energy-hungry — those used to train AI even more so. Electricity consumption, mostly from data centers, added nearly a million metric tons of pollution to the company's carbon footprint in 2023 and represents the biggest source of Google's additional emissions last year.

Full report: <https://www.gstatic.com/gumdrop/sustainability/google-2024-environmental-report.pdf>

Comments

u/handofmenoth • 1 points • 2024-07-04 04:24 UTC

I remember when Google had an official philosophy of 'don't be evil.' I also remember when they officially dropped that.

<https://gizmodo.com/google-removes-nearly-all-mentions-of-dont-be-evil-from-1826153393>

u/jesssoul • 1 points • 2024-07-04 04:05 UTC

I uninstalled Google search from my phone because of this as it gives you no option to turn off AI searching. 8.5 BILLION searches every day. Complete climate disaster.

u/Living-Situation6817 • 2 points • 2024-07-03 22:41 UTC

Data centers will need their own nuclear reactors soon...

u/[deleted] • 1 points • 2024-07-03 19:11 UTC

[removed]

u/UserInside • 1 points • 2024-07-03 19:04 UTC

As the leather jacket man said: ****"The more you buy, the more you save"****

u/[deleted] • 1 points • 2024-07-03 16:09 UTC

[removed]

u/Economy-Fee5830 • 12 points • 2024-07-03 06:37 UTC

The article does not mention that, on a global basis Google has purchased an equivalent amount of renewable electricity compared to what they used, and on average the grid they use to power their data centres are 64% renewable-powered.

However the way the Scope 2 calculations are done you have to purchase clean energy on a regional basis, and this is difficult to do in Asia, hence the increase in their Scope 2 emissions.

[See page 34](https://i.imgur.com/qpL4NJt.png). <https://www.gstatic.com/gumdrop/sustainability/google-2024-environmental-report.pdf>

Overall they still matched all the energy they used with clean energy (which was then used by someone else, cleaning up their process), and they say via their purchase agreements they encourage the building of new renewable energy projects to expand renewable grid capacity.

u/StartledWatermelon • 8 points • 2024-07-03 16:11 UTC

>they still matched all the energy they used with clean energy

No. They matched all the energy they used with dollars that were given to renewable power plants. It's better than nothing. But these dollars don't cancel the climate effects of 9 million tons of CO2 that Google released into atmosphere in 2023.

u/Economy-Fee5830 • -10 points • 2024-07-03 16:14 UTC

I know you are angry and you want to vent, but there is such a concept as **NET** Zero. Notice the net means you add things up, whether they are plus and minus and it adds up to zero.

In this case 13 million tons extra CO2 - 13 million tons worth of zero carbon energy used by someone else which comes out to 0.

So no extra carbon was released.

It's maths - look it up.

u/StartledWatermelon • 2 points • 2024-07-03 16:54 UTC

I'm not angry at all. Where did you get that?

Take a closer look at Figure 18 and you'll find out that extra carbon WAS released by Google. In fact the amount of extra carbon, i.e. one that isn't matched by zero-carbon energy purchases, has grown more than 4x since

2019.

Net zero is an economically efficient way of shifting emissions between different users. But there're ethical considerations too. And they diverge from economical benefits quite a bit. Funny thing is, big corporations love push onto public claims about their ethical virtues. Their economic drives, however, are discussed in a much more hushed tone.

u/Economy-Fee5830 • -4 points • 2024-07-03 17:29 UTC

> Take a closer look at Figure 18 and you'll find out that extra carbon WAS released by Google.

You obviously did not understand the original statement or what scope 2 means.

Scope 2 does not allow for global accounting, only regional.

Extra carbon was released in Asia, and extra carbon-free energy was purchased in Europe.

Next time read a bit more closely before commenting.

u/StartledWatermelon • 2 points • 2024-07-03 21:00 UTC

Scope 2 allows some wiggle room actually but this is not the point.

The main difference in our view boils down to, you seem to put emphasis on *accounting* and I put more focus on the actual CO2 released into atmosphere. You can guess which one impacts global climate and which one isn't.

Unfortunately, there aren't viable options for the Google to substantially lower its actual emissions.

Now, if you have chosen such condescending tone, you should be able easily rebuke this thesis, made by my colleague (most likely he didn't come with it himself but he formulated it succinctly enough):

>A rich company decarbonizes by buying green energy from a country where it's easily available, probably with generous state subsidies to producers of said energy.

>Which makes further decarbonization progressively more challenging, since it requires green energy to be bought by poor consumers in countries where green energy is hard to procure and there isn't state policies working in its favor

It'd be really appreciated if such rebuke won't contain condescending remarks.

u/Economy-Fee5830 •
0 points •
2024-07-03
21:45 UTC

> you seem to put emphasis on accounting and I put more focus on the actual CO2 released into atmosphere

It's the same thing. Dont you understand that? Are you stuck on some stupid moral high ground or

something
that you
cant see
common
sense?

Your
analogy is
also
stupid.
Firstly,
Google
says they
focus on
developing
markets
for clean
energy
and
secondly
we are far
from the
point
where the
market is
super-
saturated
with clean
energy -
all the
clean
energy
we buy
still
displaces
CO2.

> Bundled
energy
and
EACs:
Our
approach
to
clean
energy
procurement
to date
prioritizes
purchasing
physical
electricity

along with associated “bundled” EACs. 110 **This helps ensure the additionality of our clean energy projects (i.e., that they’re new to the grid)**. While the GHG Protocol allows companies to apply “unbundled” EACs to their Scope 2 footprint, including from existing clean energy projects, **our “bundled” approach prioritizes new clean energy projects, which helps drive the clean energy transition** but leaves us with some regional

gaps

Get off
your
stupid
anti-
business
high
horse.

u/ConvenientShirt • 3 points • 2024-07-03 16:45 UTC

Net zero means nothing when it only accounts for increases in energy usage. Building a solar farm to provide renewable energy is great I, but when that solar farm is only built so it can supply dedicated power to a project or offset new energy expenditures it's literally a waste of the resources needed to create the solar panels because they can never recompensate the system that spent energy to create them.

All you are doing is wasting materials meant to bring energy independence and reduce carbon emissions by straining the system to waste more carbon to produce panels that the public at large sees literally no benefit to. Google is patting itself on the back about using green energy like it just appears out of thin air, denying that renewable energies rely on finite materials and take a lot of carbon to produce, and instead of actually being used to offset energy usage in ways that take polluting power plants offline, they are being used to make Google money while the rest of our systems are still seeing increases in demand and production of non clean energy infrastructure.

The only way to solve this problem is to reduce our energy usage in general, and currently all of what we built to save us is being used as excuses to astronomically increase our overall energy usage, all of the gains we have had in producing renewable energy resources has been undone by carbon demand by AI data centers.

u/Economy-Fee5830 • -6 points • 2024-07-03 17:26 UTC

> it's literally a waste of the resources needed to create the solar panels because they can never recompensate the system that spent energy to create them.

This is a pretty stupid position since value is being created by the product. There is a net increase in value for 0 carbon emissions.

> denying that renewable energies rely on finite materials and take a lot of carbon to produce,

Again an intensely stupid talking point use by degrowthers.

> The only way to solve this problem is to reduce our energy usage in general, a

The mask comes off.

u/Cthaza • 1 points • 2024-07-03 09:50 UTC

This guy's whole profile is just stuff about country emissions, he must either be a climate scientist, work in one of the many fields that school of science pertains to, ooor (tin foil hats ready?) He was hired by some government entity somewhere.

u/Economy-Fee5830 • 1 points • 2024-07-03 09:51 UTC

> This guy's whole profile is just stuff about country emissions

Which guy?

u/Cthaza • 0 points • 2024-07-03 09:52 UTC

Yours my guy I was looking at your posts

u/Economy-Fee5830 • 4 points • 2024-07-03 09:54 UTC

If you worry about being manipulated, maybe you should consider having more information (e.g. the link to the actual Google impact report) is better than not having it.

u/Cthaza • 1 points • 2024-07-03 09:59 UTC

I mean I'm not worried about that at all I don't really care because I know the people making these charts and doing all the science behind preventing further damage to our environment are waaaaaay smarter than me and they have the best chance out of anyone to "fix" this mess

The news left it out because they'll get more attention by doing things that will make their consumers have some kind of emotional response. 99% of any kind of news company is garbage because they're all controlled by a select few groups (at least in the US)

u/Economy-Fee5830 • 1 points • 2024-07-03 10:03 UTC

I get the need for clickbait headlines - they could at least have

mentioned
it in the
body of
the
article.

u/FuturologyBot • 1 points • 2024-07-03 05:59 UTC

The following submission statement was provided by /u/Balance-:

Datacenter energy consumption is a serious concern now that big tech is ramping up AI, which seriously impact climate change. This report puts some numbers on what's happening behind the doors of one of the biggest hyperscalers.

These are very staggering numbers:

> "As we further integrate AI into our products, reducing emissions may be challenging due to increasing energy demands from the greater intensity of AI compute, and the emissions associated with the expected increases in our technical infrastructure investment," the report says. Google's data center electricity consumption alone grew by 17 percent in 2023, a "trend" it expects to continue in the future, according to the report. Already, Google estimates that its data centers accounted for up to 10 percent of global data center electricity consumption in 2023.

Please reply to OP's comment here:

https://old.reddit.com/r/Futurology/comments/1du6oco/googles_carbon_footprint_balloons_in_its_gemini/lbeh8zs/

u/Balance- • 7 points • 2024-07-03 05:32 UTC

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